TerraHopper:

Selected Theme: Sustainable Products for human wellness
Team Name:CAMspire
Team ID: AG25-1500448

Final Design Concept

CAMspire -

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Role in the Project: Team Leader, Concept Development, Project Co-ordination

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Role in the Project: Mechanical design, CAD modelling, Simulation

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Role in the Project: Market research, Human Factors analysis, Sustainibility

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Team Mentor -

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Problem Identified

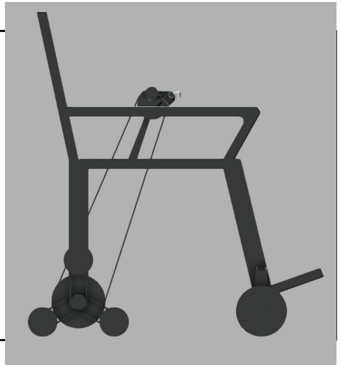
PROBLEM DESCRIPTION

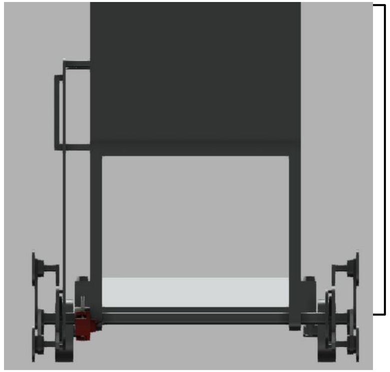
Mobility is a major challenge for elderly and differently-abled individuals, especially in regions with limited infrastructure accessibility. Existing stair-climbing wheelchairs are costly, heavy, and inaccessible to most. There is a need for a safe, affordable, and adaptive wheelchair that restores independence and enables effortless movement

Background:

1. In low- and middle-income communities, most families cannot afford advanced mobility devices. Lack of ramps, elevators, and accessible infrastructure forces elderly and differently-abled individuals to depend on others.
2. Current stair-climbing wheelchairs use track systems or robotic wheels, but they are expensive, heavy, and complex, making them inaccessible for most users in developing regions.

Representative image/video of explored ideas





Significance

WHY IS IT AN URGENT PROBLEM?

- Limited mobility traps elderly and differently-abled individuals in dependency, reducing their dignity, opportunities, and emotional well-being. Inaccessible infrastructure forces families to spend extra time, effort, and money on care, increasing financial and emotional strain.
- Solving this problem now means restoring independence, improving mental health, reducing caregiver burden, and enabling equal participation in society.

Representative image/video of explored ideas

Research findings

Representative image/video of explored ideas

What are important findings in the research?

- **What are important findings in the research?**
  Our research included surveys, interviews, and secondary data analysis.
- The findings are as follow:
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- **User Needs:** Lightweight, easy-to-use, low-maintenance, safe on stairs and flat surfaces.
- **Human Factors:** Ergonomic seating, adjustable support, intuitive controls.
- **Technology Trends:** Low-cost gear mechanisms, hybrid manual-electric systems, recycled material frames.
- **Social Context:** Limited public infrastructure accessibility; high import cost of advanced wheelchairs.

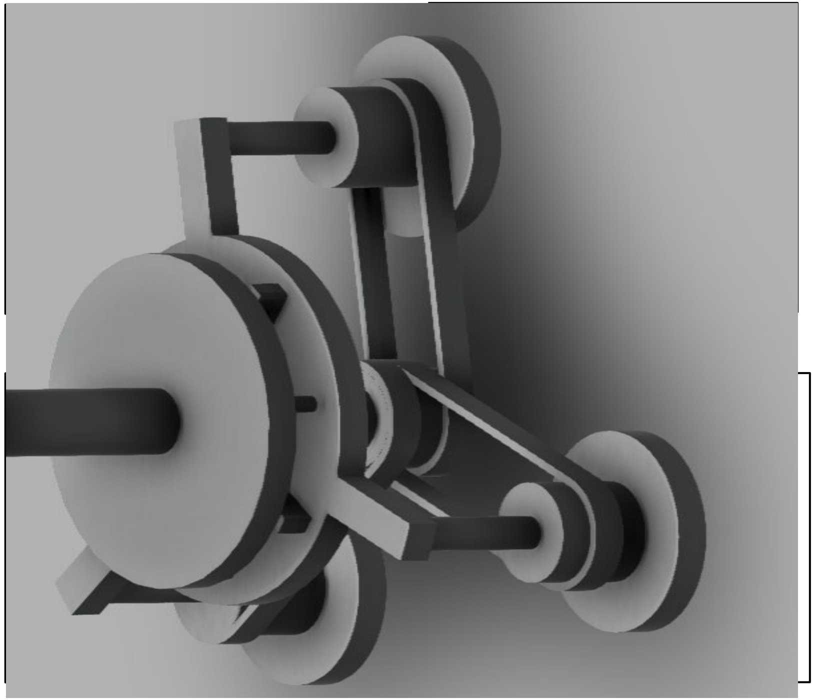
Final solution & Innovation

- Our solution uses a unique sun-planet gear system in rear wheels for efficient manual power transmission.
- Lever mechanism switches to rotate planet carrier, enabling stair climbing on uneven terrain.
- Front disc wheels are non-powered, reducing weight and complexity.
- Hydraulic disc brake on main shaft ensures safety and control.

Representative image/video of explored ideas

Final CAD Product Images

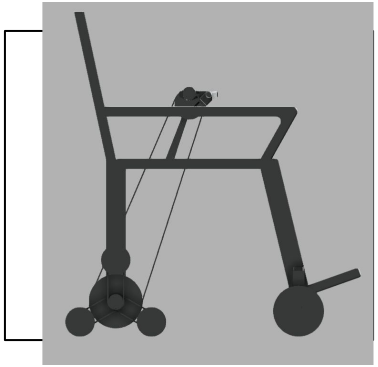
Representative image/video of explored ideas

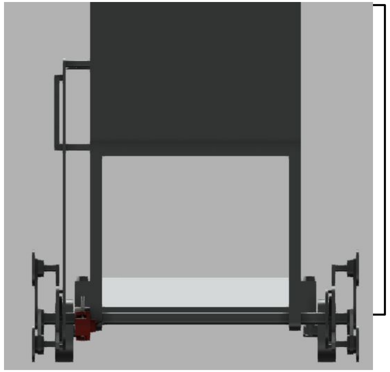


- The solution features a unique sun-planet gear system integrated into the rear wheels for power transmission. On flat surfaces, power from hand rotation drives the sun gear, which turns the planet gears, enabling smooth manual propulsion. For stair climbing or uneven terrain, a lever mechanism shifts power to rotate the entire planet carrier, allowing the wheelchair to climb efficiently. Additional innovation includes a hydraulic disc braking system on the main shaft for enhanced safety and precise control.

Product Design Specifications

Representative image/video of explored ideas

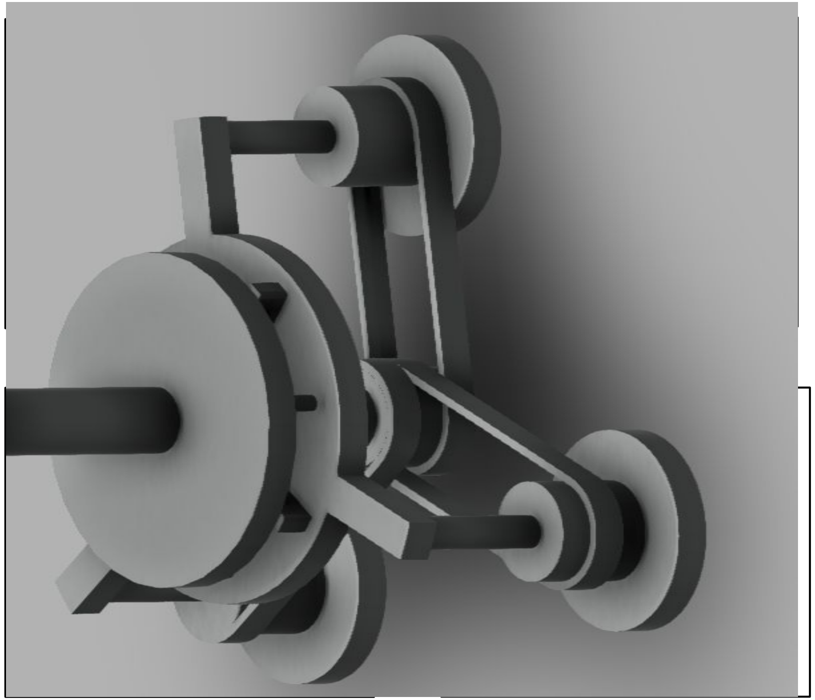





- **Components required**
  - Sun-planet gear system assembly for rear wheels power transmission.
  - Hydraulic disc brake system for effective stopping and safety.
- **Technical requirements**
  - Smooth gear engagement and reliable lever switching mechanism for mode change.
  - Durable lightweight frame supporting user weight and terrain stresses.
- **Human factors requirements**
  - Intuitive lever and brake controls for easy operation by elderly/differently-abled users.

Product Details

Representative image/video of explored ideas



The wheelchair operates on a unique mechanical system designed to provide smooth movement on flat surfaces and effective climbing on stairs or uneven terrain. The primary power transmission occurs through a sun-planet gear system integrated into the rear wheels. When moving on flat ground, the hand-rotated pulley powers the sun gear, which drives the planet gears connected to the rear wheels, enabling efficient propulsion with reduced effort.

For stair climbing or navigating uneven surfaces, a lever mechanism shifts the power transmission from the sun gear to the planet carriers, causing the entire planet gear assembly to rotate about a common axis. This rotation allows the wheelchair to climb stairs by effectively adapting the wheel movement to the obstacle.

Product Si

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Product Reflection

Representative image/video of explored ideas

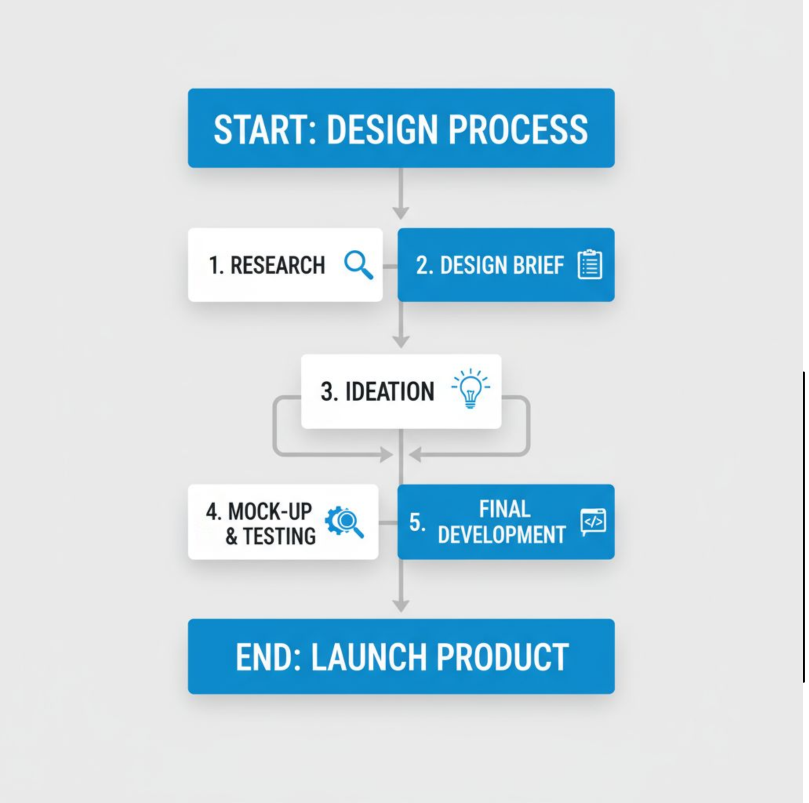
- **Benefits of your product :**
  - Much more affordable compared to heavy, imported stair-climbing wheelchairs.
  - Lightweight and portable, unlike bulky track-based systems.
  - Ergonomic and user-friendly design addresses comfort and ease of use.
  - Requires less maintenance and can be locally manufactured, unlike complex imported models.

- **Drawbacks of your products that can be optimize in future study.**
  - May have limitations in climbing very steep or irregular stairs.
  - Comfort features (like advanced cushioning or auto-adjustable seating) can be enhanced.
  - Current design may not be fully optimized for all body types and weight ranges.

Design Process

- Research: Conducted surveys, interviews, and secondary data analysis to understand user needs, infrastructure challenges, and existing solutions' limitations.
- Reframing Design Brief: Defined key specifications focusing on affordability, lightweight, ease of use, safety, and adaptability to stairs and flat surfaces.
- Ideation & Concept Selection: Explored various mechanical systems; selected the innovative sun-planet gear mechanism with lever-based mode switching for simplicity and effectiveness.

Representative image/video of explored ideas



Impact of your solution

Our solution creates a positive social impact by restoring independence and dignity to elderly and differently-abled individuals. It enables safe mobility on both stairs and flat surfaces, reducing dependency on caregivers and improving confidence.

The affordable design makes it accessible for low- and middle-income families, lowering financial strain. This leads to better mental well-being, equal participation in daily life, and reduced burden on families.

Representative image/video of explored ideas

Sustainability

Representative image/video of explored ideas

1. Promotes independence for elderly and differently-abled users.
2. Reduces caregiver burden and improves mental well-being.
3. Affordable design suitable for low- and middle-income families.
4. Uses recycled materials to minimize environmental impact.
5. Hybrid manual-electric system saves energy.

Business Viability

- Affordable Costing: Low-cost materials and production.
- Return on Investment: Saves on long-term care costs.
- Design Calculations: Lightweight frame, balanced power assist.
- Effectiveness: Smooth stairs and flat surface use.
- Efficiency: Local manufacturing, easy maintenance.

Representative image/video of explored ideas

Thank You!